

Anubhab Pradhan

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PROFESSIONAL SUMMARY

Final-year AI/Data Science student transitioning to ML systems engineering with proven expertise in building production-grade AI systems. Specialized in RAG evaluation infrastructure, statistical benchmarking methodologies, and clinical-grade ML pipelines. Demonstrated ability to architect end-to-end systems (Kairos: Go + Python + TypeScript microservices; Hasprana: production ML with ONNX quantization and clinical validation). Experienced in code auditing, failure mode analysis, and reliability engineering. Open-source contributor (LangChain). Seeking AI/ML engineering roles emphasizing system design, evaluation rigor, production optimization, and observable infrastructure.

EDUCATION

CMR Institute of Technology

Bengaluru, Karnataka

B.E. in Artificial Intelligence and Data Science | CGPA: 8.1

Expected June 2027

- Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Data Engineering, System Design

EXPERIENCE

Applied AI Engineer Intern

Jan 2026 – May 2026

Lamatic.ai

Remote

- Investigated user workflow abandonment by tracing agentic execution patterns; identified hallucinated plans, tool-routing errors, and stuck reasoning loops as primary failure modes
- Designed and validated remediation strategies across prompt engineering, routing logic, and planning algorithms using A/B testing; deployed highest-performing fixes to production
- Built multi-step agent pipelines integrating tool-use, REST/GraphQL API orchestration, and multi-hop reasoning; measured improvements via task-completion and user-retention metrics
- Implemented monitoring and debugging infrastructure to trace agent reasoning and identify failure patterns in real user workflows

AI Engineer Intern (Medical AI)

Aug 2026 – Present

Hasprana Health Care Solutions (Arohan)

Bengaluru

- Conducted comprehensive technical audit of confidence-scoring system: correlated ML softmax probabilities against segmentation error, discovering geometric confidence was statistically superior to learned confidence
- Retrained U-Net segmentation model on clinical ophthalmic data: improved mean IoU from 0.888 to 0.970 (+9.2%), reduced limbus localization error by approximately 70%
- Productionized vision pipeline with PyTorch, ONNX, and INT8 quantization; validated on 12 clinical images and 251 automated test cases achieving clinical-grade reliability
- Implemented inference optimization reducing model size by 75% while maintaining accuracy; deployed quantized model to production edge devices

PROJECTS

Kairos | RAG Evaluation & Observability Workbench

Deployed

- Engineered full-stack RAG experimentation platform combining hybrid retrieval, multi-hop reasoning, and statistical evaluation with 12+ IR metrics and confidence intervals
- Designed microservices architecture: Go gateway + Python intelligence services + Next.js portal; deployed with gRPC, PostgreSQL, ChromaDB, Prometheus, Docker, and GitHub Actions CI/CD
- Implemented experiment tracking, golden datasets, retrieval ablation workflows, cost tracking, and reproducibility features enabling systematic RAG analysis and A/B testing
- Built end-to-end observability pipeline capturing retrieval traces, chunk selections, scoring decisions, and generation inputs; enabled deep debugging of pipeline failures
- **Tech Stack:** Go, Python, TypeScript/Next.js, gRPC, PostgreSQL, Prisma, ChromaDB, BM25, Sentence Transformers, Reranking, Prometheus, OpenTelemetry, Docker, GitHub Actions

MukhdaX | *Face Image Provenance & Blockchain Verification Pipeline*

Deployed

- Built end-to-end verification pipeline: face detection (InsightFace/RetinaFace) → ArcFace representation (512-dim) → Google Lens reverse-image discovery → deterministic fingerprinting (Keccak-256) → on-chain Ethereum anchoring
- Enforced exactly-one-face constraint; extracted metadata (OpenGraph, Twitter, canonical tags) from all discovered public sources; implemented on-chain read-back verification without private keys
- Shipped dual interfaces: CLI (Python) and browser UI (FastAPI); full audit trail with explicit state reporting (blocked, failed, not_run) for all pipeline stages
- Integrated SerpApi for genuine Google Lens visual search; no hardcoded or predetermined results; deterministic SHA-256 fingerprinting of image embeddings
- **Tech Stack:** Python, FastAPI, InsightFace, RetinaFace, ArcFace, OpenCV, SerpApi, Google Lens API, Web3.py, Solidity, Ethereum Sepolia, Keccak-256

LocalBench | *Local LLM Benchmarking & Model Selection Platform*

Active Development

- Privacy-first, offline-first benchmarking for open-source LLMs via Ollama; measures quality, latency, throughput, memory consumption under real hardware constraints
- Implemented hardware-aware model selection: accuracy threshold + memory/latency budget constraints enable cost-performance tradeoff decisions (e.g., 1.5B vs 7B model selection on target hardware)
- Hardened structured generation with Pydantic validation and bounded retry logic; reproducible evaluation with immutable JSONL artifacts, full metadata capture, and version tracking
- Built CLI interface (Typer + Rich) and study assistant with Q&A and quiz generation from local models; no cloud API dependencies
- **Tech Stack:** Python, PyTorch, Hugging Face, Ollama, Pydantic, Typer, Rich, JSONL, Ollama HTTP adapter

RedOps | *Autonomous LLM Red-Teaming & Evaluation Platform*

Research Phase

- Engineered production-grade LLM evaluation framework: benchmarks safety (jailbreak resistance, prompt injection, toxicity, bias), groundedness, relevance with reproducible evaluation runs
- Built durable evaluation workflows with provider-independent execution, semantic LLM judging, metric pipelines, experiment tracking, and persistent per-item evidence for forensic analysis
- Implemented full observability: metric confidence scores, reasoning traces, token usage, cost breakdown, version metadata; idempotent persistence enabling replayable evaluation
- Designed evaluation metrics: safety scoring, hallucination detection, adversarial input crafting with template-free agent-based attack generation
- **Tech Stack:** Python, FastAPI, PostgreSQL, Docker, Redis Streams, SQLAlchemy, TypeScript, agent-based testing

RESEARCH & ACADEMIC

Co-Authored Conference Paper

ICAC2N 2026 (Published)

Benchmarking Instruction-Tuned Small Language Models on Extractive QA

- Systematic evaluation of Phi-3-mini, Mistral-7B, Gemma-2 on identical extractive QA datasets under controlled conditions
- Quantified accuracy-efficiency tradeoffs across model sizes; demonstrated metric choice determines model ranking across different evaluation methods
- Validated findings across multiple evaluation approaches to inform real-world model selection decisions

OPEN SOURCE CONTRIBUTIONS

LangChain – PR #31802 (Merged): Fixed missing `else` branch in `_evaluate_in_project()` method

LangChain – PR #38465 (Submitted): `KeyError` fix in `file_tool` callback for rename operations; comprehensive test coverage

Metaflow – Workflow optimization contributions to distributed ML pipeline orchestration

AOSSIE/OpenVerifiableLLM – LLM verification framework improvements

CO-CURRICULAR ACTIVITIES & COMPETITIONS

Hackathons: Smart India Hackathon 2024, Gen AI Exchange 2025, EY Tecathon 6.0

Team Leadership: ML Lead, Data Voyagers (university ML club); coordinated technical strategy and mentored teammates on model selection and data preprocessing

Upcoming: Hacker House Goa 2026 (preparation underway) – Focus: LangGraph and LangChain-based agentic systems

Online Certifications: Data Science Job Simulation (Forage), Introduction to Machine Learning (NPTEL-SWAYAM), Full Stack Development (Infosys Springboard)

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, SQL, Java, Go (basics), C/C++

LLM & Agentic Systems: Claude API, Groq, Ollama, LangChain, LangGraph, multi-agent orchestration, tool-use patterns, streaming inference, prompt engineering

Machine Learning: PyTorch, scikit-learn, XGBoost, SHAP, model evaluation, statistical testing, computer vision (OpenCV, InsightFace, RetinaFace, ArcFace)

Production ML: ONNX inference, INT8/FP16 quantization, model versioning, A/B testing, inference optimization, hardware-aware selection

Data & Databases: PostgreSQL, ChromaDB, vector search (FAISS, embeddings), Redis, Celery, JSONL, MongoDB

Full-Stack & DevOps: FastAPI, Flask, React/Next.js, Docker, Kubernetes (basics), Git/GitHub, CI/CD (GitHub Actions, TravisCI)

Cloud & Deployment: AWS (basics), Vercel, Docker containers, orchestration

Specializations: Code audit & architecture analysis, system design, reliability engineering, evaluation methodology, reproducibility, observability

KEY DIFFERENTIATORS

System Design: Full-stack ownership of multi-service architectures (microservices, gRPC, container orchestration, observability). Kairos spans Go gateway + Python services + TypeScript frontend with production-grade infrastructure.

Evaluation Rigor: Statistical evaluation methodology, reproducible experiments, systematic metric design. Kairos implements 12+ IR metrics with confidence intervals; RedOps designs comprehensive LLM safety evaluations.

Production ML: End-to-end ownership from research prototype to deployment. Hasprana work: retraining, quantization (INT8), inference optimization, clinical validation on real medical data.

Code Audit & Architecture: Large-scale codebase analysis, architectural debt assessment, refactoring roadmapping. Audit expertise: 62K+ LOC analysis, god node detection, coupling analysis, complexity assessment.

Agentic AI: Multi-agent orchestration, tool integration, reasoning chains, failure analysis and remediation. Lamatic work: agent execution tracing, failure mode identification, workflow optimization.

CONTACT & LINKS

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